

Arpan Pathak

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Visa: Requires H1B Transfer sponsorship

SUMMARY

Systems Software Engineer with 8+ years architecting high-performance, low-latency systems at **Amazon**, **Microsoft**, and **Oracle**, now applying systems depth to Android engineering. Deep expertise in **Kotlin**, **C++**, **Rust**, lock-free concurrency, SDK design, and real-time streaming at global scale. Brings memory-safe, cache-aware, performance-profiled engineering to mobile -- the same discipline that shipped multi-million-dollar infrastructure. Currently building a **Kotlin Multiplatform** companion app for an on-device AI perception system targeting intersection safety, lane discipline, and cooperative hazard detection.

PERSONAL PROJECTS

Driving-CivicSense (Edge AI Vision) | github.com/arpanpathak/driving-civicsense-vision-model | 2026

- Edge-AI perception system built in **Rust** for intersection discipline and lane-awareness. Fine-tuning **YOLOv8/v11** and custom **CNN** architectures for stop sign compliance detection, intersection occupancy violations, and left-lane camping classification. Experimenting with model pruning and quantization for on-device inference.
- Achieving low-latency inference on low-powered single-board computers (Raspberry Pi 5 + Hailo-8L, Qualcomm AR1). Training pipeline on cloud GPU infrastructure with synthetic data augmentation.
- Plan to publish a public driving behavior dataset on **Hugging Face** to support reproducible research in traffic civility and edge-AI perception.
- 3D printing custom glasses frame in progress for aftermarket AR clip-on form factor.
- **KMP Companion App (In Progress)**: Building a **Kotlin Multiplatform** mobile app to display real-time detection alerts from the Rust pipeline via gRPC bridge. Shared business logic in KMP with Jetpack Compose (Android) and SwiftUI (iOS) frontends.

EXPERIENCE

Software Developer II Microsoft - Redmond, WA June 2025 - Present

- **C++ to Rust Migration**: Spearheaded migration of the Information Protection service from **C++** to **Rust**, shifting memory safety to compile-time. Eliminated use-after-free and buffer-overflow CVEs in the audited path while maintaining full performance parity -- a critical win for DORA compliance. Memory-safe concurrency discipline directly applies to Android platform development.
- **AI-Powered Policy Engine**: Built an AI-driven engine that auto-generates Cilium network policies and scans for misconfigurations, reducing manual policy review cycles from **4 hours** to near-real-time with zero policy drift across sovereign regions. Reactive streaming design transferable to Android lifecycle-aware architectures.
- **eBPF & Kernel-Level Networking**: Designed and deployed eBPF-based network policies using **Cilium** for Microsoft Purview's sovereign cloud regions. Authored custom **Kubernetes Operators** (Kubebuilder) to reconcile compliance states across AKS clusters.
- **Developer Workflows**: Established CI/CD pipelines, containerized microservices, and automated validation frameworks ensuring reliable multi-region releases.
- Infra: **C++, Rust, Kotlin, Python, Kubernetes (AKS), Kubebuilder, Docker, CI/CD, Linux**

Senior Software Engineer Oracle Cloud Infrastructure - Seattle, WA Oct 2024 - Mar 2025

- **High-Performance Engine**: Architected low-latency internals for a proprietary database engine, optimizing **lock-free data structures** to reduce contention in high-concurrency read-write paths, improving transaction throughput by **45,000 ops/s**. Lock-free patterns transfer directly to Android UI threading and render-performance engineering.
- **Infrastructure-as-Code SDK**: Engineered Terraform provider (developer-facing SDK) for Oracle Autonomous Database, replacing legacy control plane infrastructure with projected **\$1.2M/month cost savings**. SDK design patterns directly applicable to Android library and framework development.
- **Security & Key Management**: Integrated OCI Security Vault across control plane services for secure secret management and automated key rotation protocols.
- Infra: **C++, Rust, Kotlin, Go, Terraform, OCI, PostgreSQL internals**

Software Development Engineer II Amazon - Seattle, WA & Hyderabad, India Mar 2021 - Oct 2024

- **Real-Time ML Ranking Engine**: Architected an **ensemble model pipeline combining XGBoost and Deep Learning** for real-time ads & deal ranking across **FreeVee, Twitch, MiniTV, Prime Video, and Amazon Retail**,

driving **5% CTR increase**, **30%** higher deal impressions, and **\$5.4M monthly revenue growth** -- processing millions of events with **100ms latency**. Same latency budgeting and performance profiling required for smooth mobile UI.

- **SDK & Distributed Systems Design:** Designed a uniqueness constraint indexing solution and **SDK** for a multi-tenant NoSQL datastore using two-phase commit, achieving **serializable isolation** at 3K writes/second -- a developer-facing library used across multiple publisher teams. API design and client library architecture analogous to Android SDK authorship.
- **Conversational AI (BERT):** Developed a **BERT-based AI model** handling **5,000 QPS** at peak, reducing support ticket triage SLA from **7 days** to **2 hours**. Threading, resource pooling, and latency optimization at scale.
- **CI/CD & Developer Productivity:** Led migration to React.js microfrontend architecture with automated CI/CD pipelines, reducing deployment cycles from **5 days** to **30 minutes** -- enabling full team autonomy and faster iteration.
- **Big Data Pipelines:** Engineered **Apache Spark** jobs processing millions of daily events into Amazon Redshift for OLAP with automated dashboards.
- Infra: **Python, C++, Kotlin, Java, Spark, Kafka, DynamoDB, Redshift, SageMaker, AWS, Docker**

Software Development Engineer Razorpay - Bengaluru, India Aug 2020 - Feb 2021

- Built UPI Payment Gateway Aggregator using **Golang** and Protobuf, processing **1M+ hourly transactions**.
- Maintained **99.99% uptime** with **150ms P99 latency** under **50,000 TPM** peak loads.
- Infra: **Go, Kafka, Redis, PostgreSQL**

Software Engineer Mindfire Solutions - Bhubaneswar, India Aug 2018 - Feb 2020

- Developed full-stack workflow tools and **AR-based product search microservice**, serving 10,000+ daily queries. Mobile-adjacent engineering with camera and AR integration relevant to Android CameraX and ARCore.
- Built administrative dashboards reducing data retrieval latency from 5 min to 1 min.
- Infra: **Java, Kotlin, React, Python, MongoDB**

SKILLS

Android & Mobile: Android SDK, Jetpack Compose, Kotlin Multiplatform (KMP), MVVM, Coroutines & Flow, gRPC mobile, BLE, SQLDelight, Mobile CI/CD

Languages: Kotlin, C++, Rust, Java, Go, Python, TypeScript, C#, Shell Script

Performance Engineering: Lock-Free Data Structures, Concurrency & Threading, Memory Profiling, Cache Optimization, Low-Latency Systems, Real-Time Pipelines, GPU-Accelerated Computing, eBPF

SDK & Developer Tools: SDK Design, API Development, Developer-Facing Libraries, Terraform Providers, CI/CD, Automated Testing, Gradle (KMP), Packaging

Machine Learning & AI: PyTorch, TensorFlow, XGBoost, Transformer Models (BERT), Computer Vision (YOLOv8/v11), ONNX Runtime, On-Device Edge AI

Distributed Systems: AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS), Kubernetes (AKS, EKS), Docker, Terraform, Apache Kafka, Apache Spark, Redshift

Core CS: Data Structures & Algorithms, System Design, Networking (TCP/IP, TLS, DNS, HTTP/3), Operating Systems, Memory Safety

EDUCATION

Bachelor of Technology - Computer Science & Engineering

RCC Institute of Information Technology - Kolkata, India | 2018